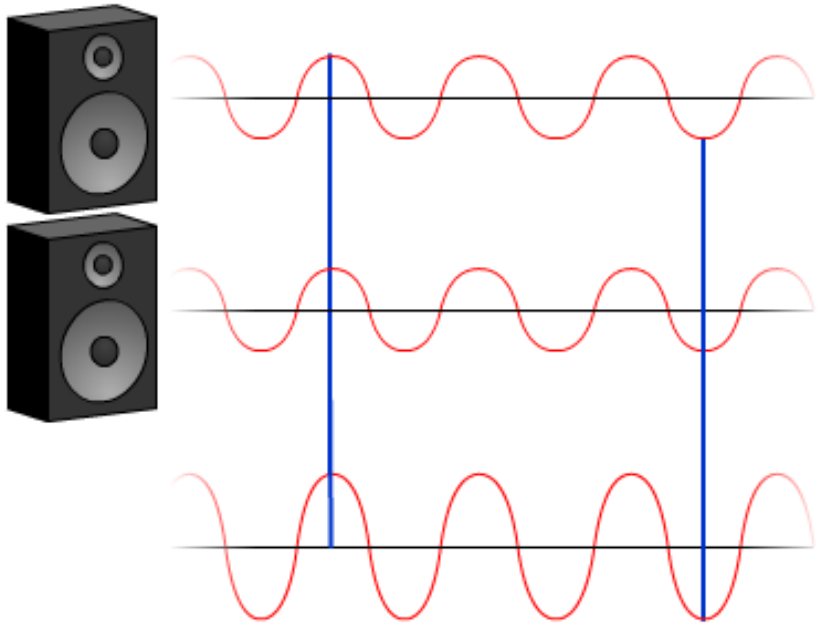
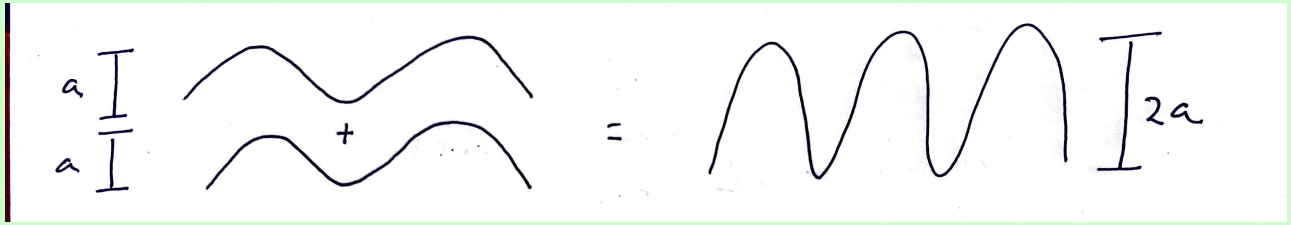
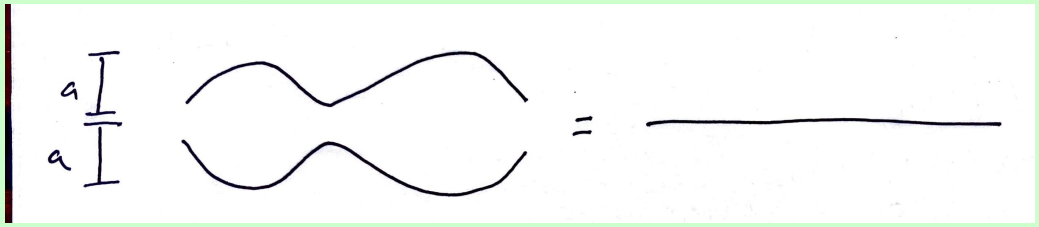
Interference



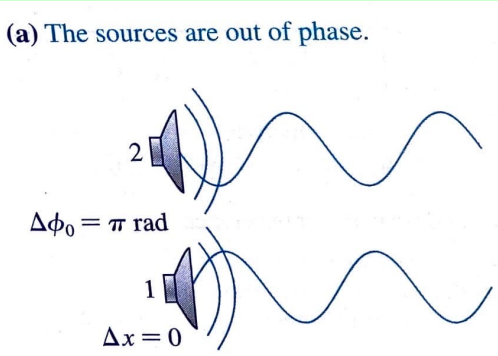
In **Constructive interference**, two waves are in phase and add

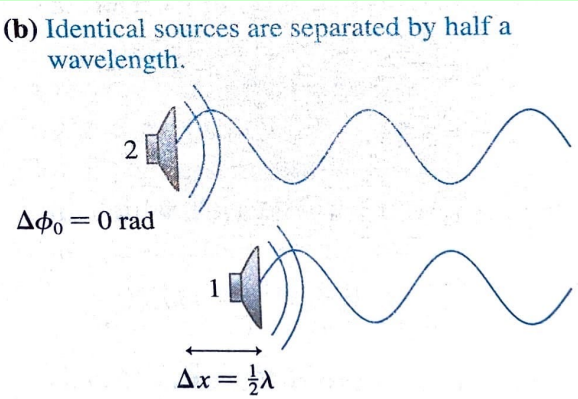


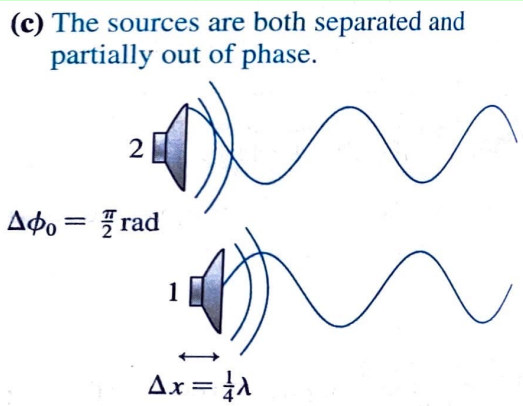
In **Destructive interference**, two waves are out of phase and cancel



There are three ways this can happen…

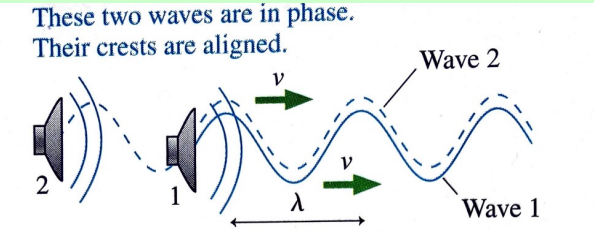


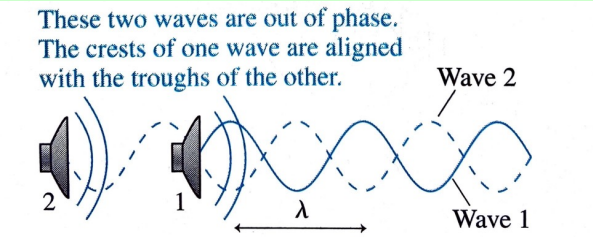




Suppose you have two speakers offset from each other, but with the same Wavelength. Its possible that the two sources could be synched up or not..

Which of these will happen at a point x1 from speaker 1 and x2 from speaker 2 ?





Define:

$$D_1 = a\sin(kx_1 - \omega t + \phi_{10}) \qquad \phi_1 = kx_1 - \omega t + \phi_{10}$$

$$D_2 = a\sin(kx_2 - \omega t + \phi_{20}) \qquad \phi_2 = kx_2 - \omega t + \phi_{20}$$

Then: $\Delta\phi = \phi_2 - \phi_1 = (kx_2 - \omega t - \phi_{20}) - (kx_1 - \omega t - \phi_{10})$

$$= k(x_2 - x_1) + (\phi_{20} - \phi_{10})$$

$$\boxed{\Delta\phi = 2\pi\frac{\Delta x}{\lambda} + \Delta\phi_0}$$

$$\boxed{\Delta\phi = 2\pi\frac{\Delta x}{\lambda} + \Delta\phi_0}$$

Where will maxima occur? $\text{If } \Delta\phi = 0, 2\pi, 4\pi \cdots = 2\pi m$

Where will minima occur? $\text{If } \Delta\phi = \pi, 3\pi, 5\pi \cdots = 2\pi(m + \frac{1}{2})$

If the two sources are identical ("coherent"), then $\Delta\phi_0 = 0$, and we can simplify

For Constructive Interference:

$$\Delta\phi = 2\pi\frac{\Delta x}{\lambda} + \Delta\phi_0 = 2\pi m$$

$$\frac{\Delta x}{\lambda} = m$$

$$\boxed{\Delta x = m\lambda}$$

For Destructive Interference:

$$\Delta\phi = 2\pi\frac{\Delta x}{\lambda} + \Delta\phi_0 = 2\pi\left(m + \frac{1}{2}\right)$$

$$\frac{\Delta x}{\lambda} = \left(m + \frac{1}{2}\right)$$

$$\boxed{\Delta x = \left(m + \frac{1}{2}\right)\lambda}$$

Finally, using trig identities it is possible to show that the amplitude of the wave Produced at any phase difference is:

$$A = 2a\cos\left(\frac{\Delta\phi}{2}\right)$$

Here is how you should solve 1D interference problems.

1. Think in terms of phase difference. This always works

$$\Delta\phi = 2\pi\frac{\Delta x}{\lambda} + \Delta\phi_0$$

If $\Delta\phi = 2\pi m$, then Constructive Interference
If $\Delta\phi = 2\pi(m + \frac{1}{2})$, then destructive interference

2. If the sources are identical it simplifies things, and you can think of path difference

Constructive $\Delta x = m\lambda$

Destructive $\Delta x = \left(m + \frac{1}{2}\right)\lambda$

Ex. Two speakers emitting identical wavelengths sit side by side. Initially no sound is heard. One speaker is then moved slowly away until max intensity is reached at 0.75 m.

A. What is $\Delta\phi_0$?

B. What is the wavelength of the sound waves?

C. What is the Phase Difference between the sources at $\Delta x = 0.3$ m?

D. Amplitude of the wave at $\Delta x = 0.3$ m?

Ex. Two identical sources produce a sound with a wavelength of 1.8 m. Assume that initially they are side-by side some distance away from you.

A. If you are 2 m away, what do you hear? Max or min?

B. Now one source is moved away. At what distance from you do you first hear a min?

C. When do you hear another max? Another min?

D. The second source is placed exactly 2.3 m away from you. What fraction of the full intensity is the sound?

Ex. 500 nm identical laser sources are initially in phase and are side by side. How far should you move one for first minimum? The next maximum?

Ex. Two loudspeakers have a maximum intensity when separated by $\Delta x = 20$ cm and have a minimum intensity when separated by $\Delta x = 60$ cm.

A. What is the wavelength of the source?

B. What was the initial phase difference between sources, $\Delta\phi_0$?


Interference Demo: Speakers
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